BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Semester Final | Semester: Fall 2025
Duration: 2 hours | Full Marks: 55

CSE 320/EEE361/ECE361: Data Communications

Answer the following questions.
Figures in the right margin indicate marks.

SET A

Name:	ID:	Section:

1. [CO2]

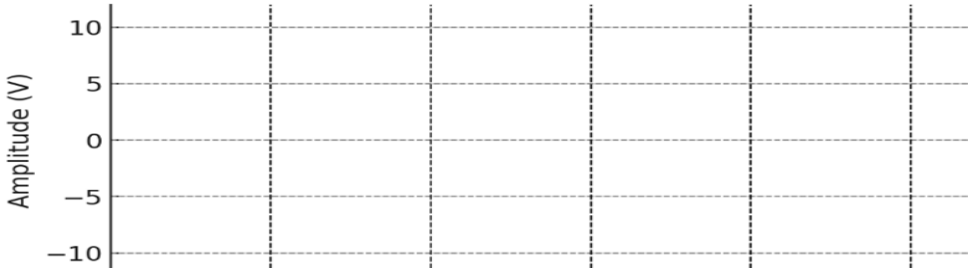
a) A 4-FSK communication system has the following parameters: [1+2+2]

- Carrier frequency f_c = 5 Hz
- Frequency deviation Δf = 2 Hz
- Amplitude: 5V

i. **Determine** the number of bits per signal.

ii. **Find** the frequencies used for the 4-FSK (M-FSK), if the frequencies are determined by the formula $f_k=f_c+(k-(M+1)/2)\Delta f$, where k=1,2,3,4 and M is the number of frequencies **[Hint: k=1 refers to 00, k=2 refers to 01 and so on]**

iii. **Draw** the transmitted FSK signal in the graph given below for the following data: **10000100**



b) Explain why digital data must be converted into analog signals for certain transmission media. What role does the carrier signal play in this process? [3]

c) Briefly discuss why Phase Shift Keying (PSK) is more robust against noise compared to Amplitude Shift Keying (ASK). [2]

2. [CO3]	a)	**Six** analog signals are multiplexed using **Frequency-Division Multiplexing (FDM)** to transmit over a single communication link. The bandwidth requirements of the channels are as follows: • Channels 1–4: 80 kHz each • Channels 5–6: 120 kHz each A guard band of 15 kHz is inserted between every pair of adjacent channels. i. **Determine** the total bandwidth required for the multiplexed FDM signal if the initial frequency starts from 301 kHz. ii. **Draw** the spectrum of the FDM signal. iii. Suppose the guard band is reduced from 15 kHz to 10 kHz. **Briefly explain** the impact on the received signal and the data rate.	[1+1 +2]
	b)	Consider a system with 12 source channels, each generating data at a rate of 2000 bits per second. The channels are combined using **Statistical Time Division Multiplexing (TDM)** with character interleaving. Since not all sources are active at the same time, each frame carries data from 8 active sources. Each slot in the frame requires an additional 3-bit address field to identify the source. i. **Determine** the size of the output frame in bits. ii. In which scenario, synchronous TDM is efficient over statistical TDM. Briefly justify your answer.	[2+2]
	c)	A Time-Division Multiplexing (TDM) system consists of m input lines. The first two input lines operate at a data rate of 16 kbps each, the third input line operates at 29 kbps, and all remaining input lines operate at 32 kbps each. The interleaving unit is 2 bits. In addition, synchronization overheads of 2 bits per frame is used. The output data rate is 288 kbps. Under these constraints, i. **Draw** the figure to visualize the multiplexing process with proper labels. ii. **Calculate** the input slot duration. iii. **Find** the Frame rate. iv. **Determine** the Frame size. v. **Identify** the number of input lines that can be supported.	[2+1 +1+1 +2]

3. [CO4]

a) A sender transmits the code word 100100101. Using the notation **C(9,6)**, **find** the corresponding **divisor.** If Cyclic Redundancy Check (CRC) is used at the receiver side to detect errors, **determine** whether the received code word 101100101 contains any error. (**Hint:** In the divisor, the 2nd bit from the Most Significant Bit (MSB) is 0 and all other bits are 1.) [2+3]

b) A linear block coding scheme is used to encode Datawords into Codewords. The following table shows the encoding scheme: [1+3 +3]

Dataword	Codeword
00	0000000
01	0111100
10	1000111
11	?

Note: In a linear block coding scheme, XOR-ing two distinct valid codewords produces another distinct codeword.

- From the table above, **find** the missing codeword.
- **Determine** the maximum number of errors this scheme can detect and correct.
- The receiver received the codewords 1111111 and 0101000. **Determine** if the original dataword can be retrieved from it. If so, what are the original datawords? **Show** detailed calculations.

c) You are given the following four 16-bit Checksum calculations at a sender device, where some of the values are missing. Your job is to analyze the calculations and answer the following questions. All the words are 16-bits and represented in Hexadecimal values. [2+2]

You **must** also answer in **Hexadecimal values**. For simplicity, the fake checksum (word with all zeroes) has been omitted in the summation process.

BAD	First Data
FACE	Second Data
?	Third Data
BEEF	Fourth Data
(+)	
C048	Initial Sum
2	Sum wrapping value
(+)	
?	Wrap Sum
?	Final Checksum

		i. What is the actual value for **Warp Sum** and **checksum**? ii. What is the actual value for **Third Data**?	
4. **[CO4]**	a)	In a computer lab, there are several computers which share a common wired network. During the lab classes, all computers are used to access online resources. Each node listens to the channel, and if it finds the channel idle, it transmits the data instantly. If it finds the channel busy, it continuously senses the channel until the channel becomes idle. i. Name which CSMA variant (1 persistent, non-persistent or p-persistent) is used here. ii. Discuss a major issue with the variant mentioned in (i) iii. CSMA protocol cannot detect collisions. Name the protocol that can detect a collision. Explain what happens after a collision is detected.	[1+2 +2]
	b)	In a shared communication link, there are two stations on either side. They are using the CSMA/CD protocol to detect any collision occurring during transmission. The minimum frame size to detect collision is 600 bytes, the speed of propagation is 6×10^7 m/s, and the bandwidth of the channel is 20 Mbps. i. **Calculate** the minimum transmission time to detect a collision. ii. **Find out** the vulnerable time of the system. iii. **Identify** the distance between the two stations.	[2+2 +2]
	c)	A disaster management authority uses a **satellite-based communication system** to receive messages from **hundreds of remote emergency stations** deployed across coastal and mountainous regions. All stations share the same communication channel to send alert messages to the satellite. During earthquakes or tsunamis, emergency stations send alert messages **immediately after detection**, without waiting for any timing coordination from the satellite. Messages may start transmission at **any instant**, and collisions are common during peak emergency reporting. The frame duration is 51.2 μs and the propagation delay is 150 ms for any message transmitted. i. **Explain** with proper reasoning which technique is used in this case? ii. **Calculate** the vulnerable time for the communication system.	[2+1]

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